Uni. Roll No:- _____

End Term ODD SEMESTER Examination, 2015 – 16

B.Tech. II Year III Sem. (Common to ALL branches)

Subject: - ENGINEERING MATHEMATICS - III (AHM – 3001)

Time: - 2 Hrs. 30 Minutes **Total Marks:- 40**

Note: Attempt **ALL** sections. The terms, used in question paper, have their usual meanings.

SECTION A

(16 x 1 = 16 marks)

Instructions: This section has **16** questions. Attempt **ALL** questions of this section.

Q.1. Form the partial differential equation from the relation :

$$z = a x + by + a b$$

Q.2. Solve the partial differential equation :

$$p x + q y = z .$$

Q.3. Solve the partial differential equation :

$$r - 4s + 4t = 0.$$

Q.4. Find the particular integral of the partial differential equation:

$$r + 2s + t = 2 (y - x).$$

Q.5. Solve the partial differential equation :

$$r + 2s + t + 2p + 2q + z = 0$$

Q.6. Classify the partial differential equation :

$$u_{xx} + u_{yy} + u_{zz} = 0.$$

Q.7. Find the Laplace transform of $F(t) = 1^t$

Q.8. Find the Laplace transform of

$$F(t) = \lfloor 2$$

Q.9. Find the Laplace transform of

$$F(t) = t \sin t$$

Q.10. Find the inverse Laplace transform of

$$f(p) = \frac{\lfloor 2016}{p^{2016}}$$

Q.11. Find the inverse Laplace transform of

$$f(p) = \frac{1}{p^2 + 2p + 2}$$

Q.12. Find the inverse Laplace transform of

$$f(p) = \cot^{-1} p$$

Q.13. Prove that $u = e^x \cos y$ is a harmonic function.

Q.14. What are Cauchy – Riemann equations in Polar Coordinates?

Q.15. Prove that :

$$\oint_c \frac{dz}{z-a} = 2\pi i \qquad \text{where c is a circle } |z-a| = r$$

Q.16. What is the nature of singularity of the function

$$f(z) = \sin\left(\frac{1}{z-a}\right) \text{ at } z = a \text{ ?}$$

SECTION B

(4 x 3 = 12 marks)

Instructions: This section has 5 questions. Attempt **ANY FOUR** questions of this section.

Q.1. If f (z) = u + iv is an analytic function, prove that u and v will be harmonic functions.

Q.2. If $v = \tan^{-1}(y / x)$ is the imaginary part of an analytic function f(z) = u + iv then find f (z) in terms of z.

Q.3. Evaluate: $\oint_c \frac{z+4}{z^2+2z+5} dz$ where c is

(i) $|z-i| = 1$

(ii) $|z-1-i| = 2$.

Q.4. Prove that an analytic function w = u + iv cannot be expressed as a function of $\bar{z}$.

Q.5. Evaluate using contour integration:

$$\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta} \quad ; \quad a > |b| > 0$$

SECTION C

(3 x 4 = 12 marks)

Instructions: This section has 4 questions. Attempt **ANY THREE** questions of this section.

Q.1. If f (z) = u + iv is a regular function of z, prove that :

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4|f'(z)|^2$$

Q.2. Evaluate $\int_0^{1+i}(x-y+ix^2)dz$ along:

(i) the straight line y = x

(ii) the real axis from z = 0 to z = 1 and then along a line x = 1 from z = 1 to z = 1+ i.

Q.3. Expand $f(z) = \frac{1}{z^2-3z+2}$ in the Taylor or Laurent's series in the following regions:

(i) $|z| < 1$

(ii) $|z| > 2$

(iii) $0 < |z-1| < 1$

Q.4. Find the residue of

$$f(z) = \frac{z^3}{(z-1)^4(z-2)(z-3)}$$ at its poles.

Hence evaluate $\oint_c f(z)dz$, using Cauchy's residue theorem, where c is the circle $|z| = 4$

Printed Pages-3 University Roll No.....

End Term Examination, Odd Semester 2015-16

CSE3001: Data Structures & algorithms

Time: 02 ½ Hours Total Marks: 40

Instruction:

1. Attempt all questions from Section-A, any four from Section-B and any three from Section-C.
2. Symbols have their usual meanings.

Section A

Note: Attempt All Questions **(1x16=16)**

I. List out the areas in which data structures are applied extensively?

II. If the characters 'P', 'Q', 'R', 'S' are placed in a stack (in that order), and then removed one at a time, in what order will they be removed?

III. How many null nodes will a binary tree contains with 20 nodes.

IV. There are 8, 15, 13, 14 nodes were there in 4 different trees. Which of them could have formed a full binary tree?

V. Write down the two applications in computer science of stack data structure.

VI. In an AVL tree, at what condition the balancing is to be done?

VII. Draw all possible nonsimilar trees where T is a binary tree with 3 nodes.

VIII. What is the complexity to search an element in worst case in BST.

IX. Write down the application areas where linked list is useful.

X. Does the minimum spanning tree of a graph give the shortest distance between any 2 specified nodes?

XI. Whether Linked List is linear or Non-linear data structure?

XII. What is the complexity of quick sort algorithm?

XIII. In a selection sort of n elements, how many times is the swap function called in the complete execution of the algorithm?

XIV. What is hashing?

XV. Write down the name of data structures used in DFS and BFS.

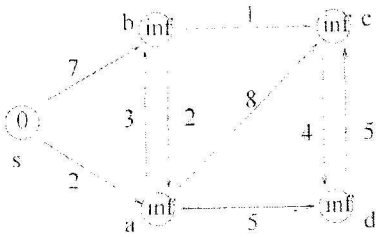
XVI. What should be the depth of a complete binary tree with n elements?

Section B

Note: Attempt any Four Questions **(3x4=12)**

I. Explain the structure operations and disadvantages of sequential files.

II. Write down the steps of Dijkstra' algorithms and implement on the following graph



III. Enlist the basic differences between tree and graph. Discuss different technique for storing graph.

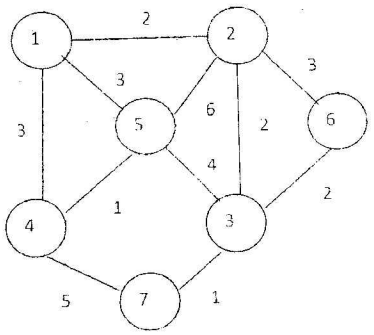
IV. What is linear probing? How it differ from quadratic probing?

V. What is Breadth first Search traversal? Explain its algorithm with an example.

Section C

Note: Attempt any Three Questions **(4x3=12)**

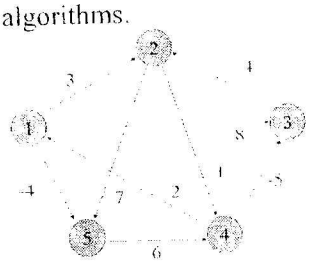
I. What is spanning tree? When it can be called a minimum spanning tree? Find Minimum spanning tree for the following Graph (G) by using Kruskal's Algorithm.



II. Using a hash table with eleven slots indexed through 0 to 10 and the hashing function h(k)= k mod 11. show the hash table that results when the following keys are inserted in the order given:
26. 42, 5, 44, 92, 59. 40, 36, 12, 60, 80.
Assume that collisions are resolved using synonyms chaining.

III. How insertions and deletions of records are handled in indexed sequential files.

IV. Compute the shortest- path between using Floyd-Warshall algorithms.



End Term Examination, 2015-16

Title of Paper: Computer Organization **Paper Code: CSE3002**

Time: 150 Min. **Max Marks: 40**

SECTION A

Note: Answer all questions. **1x16=16Marks**

I. Why is the two's complement notation for integers used in computers, in preference to other systems for coding integers?

II. Estimate how many Full adders, half adders and AND gates will be required to multiply n×n array multiplier?

III. Encode 1001 using the Hamming code and even parity.

IV. The four bit two's complement notation for the decimal integer 7 is 0111. Obtain the four bit two's complement notation for the decimal integer -7

V. The gray code equivalent of (10111) is…

VI. Decimal equivalent of 8-bit floating point number E7 (in hex) is..

VII. Which register is used to hold the data or instruction when either reading or writing?

VIII. What is the overall function of the Control Unit?

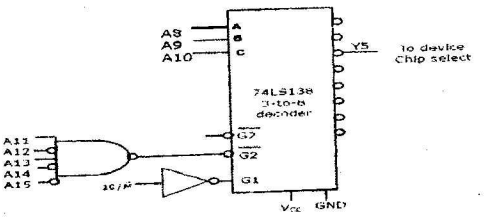
IX. What are the advantages of hardwired control unit? What are the disadvantages?

X. What do you mean by Data Hazards?

XI. What are the different types of shift register?

XII. Explain the flip-flop excitation tables for JK FF.

XIII. In the circuit shown, the device connected to Y5 can have address in the range…



XIV. How many address lines are required in a 4096 x 8 EPROM CHIP?

XV. An 8 bit microprocessor based system uses a 4K x8 bit RAM whose starting address is AA00H. The address of the last byte in this RAM is...

XVI. Differentiate between sequential access and random access memory.

SECTION B

Note: Answer any four questions. **3x 4= 12 Marks**

I. The computer has direct map cached with 16 one word blocks. The catche is initially empty. what is the observed hit ratio when the CPU generates the following word address sequence:

1, 4, 8, 5, 20, 17, 19, 56, 9, 11, 4, 43, 5, 6, 9, 17.

II. In two level memory system,the average access time without Level 1 is 150nSec and with level 1 is 30 nSec.

If level 1 access time is 20 nSec,Then:

i. What is the Hit Ratio
ii. If Hit ratio is made to 100%, what will be the access time of Level 1 and Level 2 memory?
iii. If the average access time is increased by 10%, then what will be the % change in Hit Ratio.

III. Interface two numbers of 8kb EPROM & one number of 4kb RAM with 8 bit microprocessor system having 16 bit address. Draw the interface diagram and allocate binary address to memory ICs.

IV. Write short notes on

a. EEPROM

b. Flash memory

c. SRAM and DRAM

V. (a) Differentiate between the memories mapped I/O and peripheral mapped I/O?

(b)What is Interrupt? How Vectored interrupt is different from non vectored interrupt

SECTION C

Note: Answer any three questions. **4x 3=12 Marks**

I. Consider a machine with a byte addressable main memory of 216 bytes and block size of 8 bytes. Assume that a direct mapped cache consisting of 32 lines is used with this machine.

i. How is a 16-bit memory address divided into tag, line number, and byte number?

ii. Into what line would bytes with each of the following addresses be stored?

0001 0001 0001 1011
1100 0011 0011 0100
1101 0000 0001 1101
1010 1010 1010 1010

iii. Suppose the byte with address 0001 1010 0001 1010 is stored in the cache. What are the addresses of the other bytes stored along with it?

II. What is virtual memory? Explain address translation mechanism for converting virtual address into physical address with neat diagram.

III. Consider a memory system that uses a 32-bit address to address at the byte level, plus a cache that uses a 64-byte line size.

i. Assume a direct mapped cache with a tag field in the address of 20 bits. Show the address format and determine the following parameters: number of addressable units, number of blocks in main memory, number of lines in cache, size of tag.

ii. Assume an associative cache. Show the address format and determine the following parameters: number of addressable units, number of blocks in main memory, number of lines in cache, size of tag.

IV. What is DMA? Explain DMA operation with a diagram. Also explain data transfer modes in DMA.

END Term Examination, 2015-16
Course: B. Tech (CS), Year: II, Semester: III

Database Management System (CSE3003)

Uni. Roll No:
Time: 2.5 Hrs. **Total Marks: 40**

Notes:
1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions. 1 x 16 = 16 marks

I. What are five main functions of a database administrator?
II. Explain the distinctions among the terms primary key, candidate key?
III. List two reasons why we may choose to define a view.
IV. Why certain functional dependencies are called *trivial* functional dependencies?
V. Why ordering of tuple are not allowed in relation.
VI. What is the total participation?
VII. Define BCNF normal form.
VIII. List the three design goals for relational databases.
IX. Consider the given relation R (ABCDE) with the set of the following function dependencies: F = {A→BC, B→D, CD→E, E→A} Find the closure of the AED.
X. What are the two condition of union compatibility?
XI. What do you understand by spurious tuple.
XII. Define the meaning of aborted transaction.
XIII. What do you understand by serial schedule?
XIV. Define distributed database.
XV. What do you understand by incorrect summary problem?
XVI. Suppose that there is a database system that never fails. Is a recovery manager required for this system?

Section –B

Note: Attempt Any Four Questions. 3 x 4 = 12 marks

(I) Explain the usefulness of the transaction processing system properties?

(II) Database-system implementers have paid much more attention to the ACID properties than have file-system implementers. Why might this be the case?

(III) What do you understand by horizontal and vertical fragmentation?

(IV) When two schedule are considered to be equivalent? Explain with the help of suitable example.

(V) Explain how the following differ: fragmentation transparency, replication transparency, and location transparency.

Section –C

Note: Attempt Any Three Questions. 4 x 3 = 12 marks

(I) **A)** Consider the following two transactions:

```
T1: read(A);
    read(B);
    if A = 0then B := B + 1;
    write(B).
T2: read(B);
    read(A);
    if B = 0then A := A + 1;
    write(A).
```

Let the consistency requirement be $A = 0 \vee B = 0$, with $A = B = 0$ the initial values.

a. Show that every serial execution involving these two transactions preserves the consistency of the database.

b. Show a concurrent execution of T1 and T2 that produces a nonserializable schedule.

c. Is there a concurrent execution of T1 and T2 that produces a serializable schedule?

B) Discuss the relative advantages of centralized and distributed databases.

(II) **A)** Consider the following two transactions:

T31: read(A);
read(B);
if A = 0then B := B + 1;
write(B).

T32: read(B);
read(A);
if B = 0then A := A + 1;
write(A).

Add lock and unlock instructions to transactions T31 and T32, so that they observe the two-phase locking protocol. Can the execution of these transactions result in a deadlock?

B) How might a distributed database designed for a local-area network differ from one designed for a wide-area network?

(III) Consider a relation that is fragmented horizontally by *plant-number*: *employee* (*name, address, salary, plant-number*) Assume that each fragment has two replicas: one stored at the New York site and one stored locally at the plant site. Describe a good processing strategy for the following queries entered at the San Jose site.

a. Find all employees at the Boca plant.

b. Find the average salary of all employees.

c. Find the highest-paid employee at each of the following sites: Toronto, Edmonton,Vancouver, Montreal.

d. Find the lowest-paid employee in the company.

(IV) During its execution, a transaction passes through several states, until it finally commits or aborts. List all possible sequences of states through which a transaction may pass. Explain why each state transition may occur.

University Roll No.....

B.Tech. II[nd] Year, III Semester, End Term Examination(2015-16)

MBA-5031: Industrial Economics

Time: 150 Mins. **Total Marks: 40**

Section –A

Attempt All Questions **1×16=16 Marks**

1. Define Industrial Economics.
2. Write down basic economic problems?
3. Write down the Scarcity Definition of Economics.
4. What do you mean by Elasticity of Demand?
5. Write down Law of Demand?
6. What do you mean by the term "Trade-Off"? With which concept of economics it is related?
7. Define the term "Isoquant".
8. Write down the concept of 'Fixed Cost' with suitable example.
9. What do you mean by economies of scale?
10. Write down the law of diminishing returns.
11. Write down factors of production with factor payments.
12. If AFC is falling throughout, will MC necessarily be falling?
13. What is Oligopoly?
14. What do you mean by Chronic Inflation?
15. Define Business Cycle.
16. Define the term "Market Structure".

Section –B

Attempt Any Four Questions Only **3×4=12 Marks**

1. What do you mean by the term 'Market'? Discuss its various characteristics.

2. Apart from Competition base classification, what are various possible market types? Briefly discuss.
3. Define Inflation. Discuss its various impacts.
4. Discuss the concept of Price Rigidity with respect to Oligopoly market.
5. Discuss various types of Inflation.

Section –C

Attempt Any Three Questions Only **4×3=12 Marks**

1. Discuss various Competition based Market Structures with their characteristics.
2. Discuss various factors causing "Inflation".
3. Define National Income. Discuss various methods of calculating National Income.
4. Discuss various stages of Business Cycle.

Set – A

University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. II Year (Odd Semester), End Term Examination, 2015-16

Subject: Business English | Subject Code: AHE 3080

Time: 90 Minutes | Max. Marks: 20

Notes:

- Answer all questions from Section A, any 3 out of 4 from Section B and Section C.
- Answers should be to-the-point, and word limit for informative questions from Section B (approx 60 words) and Section C (150 words) should be maintained.
- All questions of a particular section should be answered at one place.

Section A **0.5 x 10=5**

1. What does BCC mean in an e-mail?
2. Politeness and formality are the prominent features of:
   a. Downward communication
   b. Upward communication
   c. Horizontal communication
   d. None of these
3. What is interpersonal distance in intimate proxemic zone?
4. What phrase can be used to take a message in a telephone call?
5. Name any one barrier to effective listening.
6. What phrase/expression can be used for disagreeing in a Group Discussion?
7. Which stage of team formation is the most challenging one for the team leader?
8. The tendency of using only "He" while referring to a person in general is an instance of ..... barrier.
9. Suggest one method to overcome anger.
10. Write any two qualities of a good leader.

Section B **2x3=6**

1. Explain the HURIER Model of effective listening, in brief.
2. Identify the number of syllables in the following words and mark stress too:
   A. Rationalization B. Antinationalistic

C. Extemporaneous D. Lieutenant

3. Being assertive is more beneficial than being aggressive in negotiation. Explain with a suitable example.
4. What points should be kept in mind while drafting an e mail to a person senior in rank?

Section C **3x3=9**

1. Define barriers to communication. Describe the types of barriers; detail any three of the barriers with examples.
2. Getting inputs through listening is the first step towards language learning. Justify.
3. A). Discuss the role of initiator in GD.
   B). Do you think that the format of GD is more democratic than debate?
4. If you underperform in a corporate workplace, stress leads to frustration and frustration leads to depression. Comment.

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SET [D]

Printed Pages: 16 University Roll Number:

ODD

B. Tech & MCA II Year, III Semester, End Term Exam-2015 – 16

Ethics & Values: AHE-3001

Time: 2 ½ Hours Max. Marks: 40

Note: This paper consists of four passages. The passages are not unseen and are based on course discussion. They help in your analytical problem solving. Carefully read the passages and answer the following questions having the multiple answers. Carefully fill all the information in OMR sheet neatly in the blocks. No partial marking.
Full marks (assigned) will be given for a complete answer that means all correct choices.

Section A

Passage: 1 (1.5X8=12) (Attempt All Questions)

Ethics deal with the rightness and wrongness of the social moral systems. Ethical philosophy differs from the sciences because it is normative or prescriptive, rather than descriptive. In other words, ethics tell us how we ought to act or what we should do, while the sciences are more likely to observe how things are in nature. There are mainly three aspects of moral judgments: 1) Purpose or motive, 2) Act, rule or Maxim, and 3) Results or consequences. Utilitarianism says that the Result or the Consequence of an Act is the real measure of whether it is good or bad. This theory emphasizes Ends over Means. Theories, like this one, that emphasize the results or consequences are called teleological or consequentialist. Jeremy Bentham formulates the theory and argues that Man is under two great masters, pain and pleasure. The great good that we should seek is happiness. (a hedonistic perspective). Those actions whose results increase happiness or diminish pain are good. They have "utility." In determining the quantity of happiness that might be produced by an action, we evaluate the possible consequences by applying several values: Intensity, duration, certainty or uncertainty, propinquity or remoteness, fecundity, purity, and extent. There are four views of utilitarianism. 1) Consequentialism: The rightness of actions is determined solely by their consequences. 2) Hedonism: Utility is the degree to which an act produces pleasure. Hedonism is the thesis that pleasure or happiness is the good that we seek and that we should seek. 3) Maximalism: A right action produces the greatest good consequences and the least bad. 4) Universalism: The consequences to be considered are those of everyone affected, and everyone equally. There are two formulations of utilitarian theory. One, Principle of Utility: The best action is that which produces the greatest happiness and/or reduces pain. Two, Greatest Happiness: We ought to do that which produces the greatest happiness and least pain for the greatest number of people. There are two types of utilitarianism: 1) Act:

An Action is right if and only if it produces the greatest balance of pleasure over pain for the greatest number. Bentham was the profounder of the view. 2) Rule: An action is right if and only if it conforms to a set of rules the general acceptance of which would produce the greatest balance of pleasure over pain for the greatest number. John Stuart Mill defended the view in his work on utilitarianism. Let's take some examples to understand the application of utilitarian theory. A) You attempt to help an elderly man across the street. He gets across safely. Conclusion: the Act was a good act. B) You attempt to help an elderly man across the street. You stumble as you go; he is knocked into the path of a car, and is hurt. Conclusion: The Act was a bad act. If you can use eighty soldiers as a decoy in war, and thereby attack an enemy force and kill several hundred enemy soldiers that is a morally good choice even though the eighty might be lost. But, Deontology would be against of this kind of deployed. Deontology put emphasis on the intention of the action and not on the result. If lying or stealing will actually bring about more happiness and/or reduce pain, Act Utilitarianism says we should lie and steal in those cases. Bentham's theory could mean that if 10 people would be happy watching a man being eaten by wild dogs, it would be a morally good thing for the 10 men to kidnap someone (especially someone whose death would not cause grief to many others) and throw the man into a cage of wild, hungry dogs. If I am to bring the greatest happiness to the greatest number, not putting my own happiness above others, that may lead to a dilemma. I live in a neighborhood where 83% of my neighbors use drugs. I could make them most happy by helping supply them with cheap drugs, but I feel uncomfortable doing that. What should a utilitarian do? Mill answers that mankind must by this time have acquired positive beliefs as to the effects of some actions on their happiness; and the beliefs which have thus come down are the rules of morality for the multitude, and for the philosopher until he has succeeded in finding better." Mill concludes that we should always seek improvements. Mill argues that we must consider the quality of the happiness, not merely the quantity. For example, some might find happiness with drugs, alcohol and smoking. Others may find happiness working on anti-drug campaign to make the lives beautiful of the people who are suffering from drug consumption. Thus, Mill offers a balanced view of utilitarianism than Bentham. His view creates space for a higher pleasure; in which a human analyses the pleasures in terms of quality like perception, thinking, reason and judgment. Mill says this is the realization of human spirit, happiness and morality.

Questions:

1) Which of the following statements are **wrong** about ethical philosophy?
   a) It tells the way things are in nature.
   b) It tells what we should do.
   c) It is normative or prescriptive.
   d) None of these.
2) What are **the main aspects** of the **moral decision making**?
   a) The act rule or principles of any action.
   b) The results of an action.
   c) The aim or motive of an action.
   d) None of these.
3) Which of the following statements are **true** about Bentham's utilitarianism?
   a) The greatest good that we seek is maximum happiness.
   b) The greatest good that we seek is quality of happiness.
   c) Man is under two masters, pain and pleasure.
   d) All above.
4) "If lying or stealing will actually bring about more happiness and/or reduce pain, Act Utilitarianism says we should lie and steal in those cases." What can we **conclude** about Bentham's utilitarianism from this statement?
   a) He does not mind the quality of an action.
   b) He gives importance to pleasure only.
   c) He put emphasis only on the action and its result.
   d) He put emphasis on rules, principle or merit of qualitative choice.
5) "I live in a neighborhood where 83% of my neighbors use drugs. I could make them most happy by helping supply them with cheap drugs, but I feel uncomfortable doing that. What is **right thing to** do according to Mill?
   a) We should think and reason about the actions.
   b) We should not supply the drugs to make them happy.
   c) We should look for the pleasure only.
   d) We should not consider the quality of happiness.
6) "You attempt to help an elderly man across the street. He gets across safely. Conclusion: the Act was a good act. In other case, you attempt to help an elderly man across the street. You stumble (slip) as you go; he is knocked into the path of a car, and is hurt. Conclusion: The Act was a bad act."
   What we can **infer from the deontology** in these examples?
   a) Utilitarianism is wrong as it only looks at one's motive or intention.
   b) In this view what counts is the intention to help the elderly.
   c) Goodness and badness do not depend on the quality of an action but on the successful results.
   d) Your action is good if it produces good results even by following the wrong intentions.

7) Why does **Mill's view** offer a **better view on Utilitarianism**?
   a) It considers higher pleasures.
   b) It assesses pleasure in terms of thinking and judgment.
   c) It does not differentiate pleasures in terms of their qualities.
   d) Seeking quality of happiness is about the realization of human spirit.
8) If you can use eighty soldiers as a decoy (distraction) in war, and thereby attack an enemy force and kill several hundred enemy soldiers, this seems a good choice even though the eighty might be killed. What you can **conclude** from this example?
   a) From Mill's view: we should not sanction it as it is a violation of individual rights as we are using them in the war.
   b) From Bentham's view: we should sanction it as it gives maximum good.
   c) Deontology would not allow it.
   d) None of these.

Section B

Passage: 2 (1X9=9)

One should be respected on the grounds one actually deserves. In other words, if one performs good meritorious acts one must be rewarded and if one does something wrong one must be punished accordingly. It is the argument from moral desert that theorists define justice in a manner a person actually performs and deserves accordingly. However, the moral desert argument should be understood in contrast to the affirmative action theory by which a government usually acts positively in redressing the justice to benefit the underrepresented group in the areas of education, employment and business. There are three prominent views on the justice and to its distribution to the common people. Let's first talk about the John Rawls who in his a Theory of Justice advances a thought experiment what he calls veil of ignorance is actually a situation where people are in their "Original Position" in which all so called self-interest seeking individuals, who are conscious and psychologically fit actors to take decisions or make choices on the benefits and developments, have no knowledge of their arbitrary facts say race, physical disability, cast, gender and class. The facts we do not choose and decide. And, for which we cannot claim any moral credit. Now once veil goes up they have to choose and decide on the better principles of justice by which a fair distribution of resources, powers and rights can be done. Rawls argues that there can be two principles seem reasonable in such a situation: 1) Equality principle, according to

which each individual must be provided equal rights with the most extensive basic liberties compatible with similar liberties for others, and 2) Difference principle what he calls qualified equality principle that maintains that socio-economic inequalities have to arranged in a manner so that they are to be the greatest benefit of the least advantaged members of society. Here it seems that Rawls tries to balance his theory of justice as fairness with the theory of distributive justice or social justice. Before we go to understand what Rawls actually mean by his difference principle let us try o understand three basic arguments of affirmative action theory: 1) Corrective argument; according to which a government should take positive actions to redress the wrongs of past by benefiting some cast, race, gender or group who have not been given the proper resources to develop and compete, 2) Compensatory argument which points out the fact that some advantaged groups of people have to compensate for some least advantaged groups of people who have been deprived of basic resources for development, and 3) Social Representation / Diversity argument that maintains that the different variety of groups who actually make up the whole society should be given proper and proportionate socio-political-economic representation in terms of education, job and business. However, Rawls in his qualified equality principle argues that socio-economic inequalities in terms of government affirmative action policies and provisions must be in favor of least well-off. The second prominent view in contrast to distributive theory is Aristotelian Virtue theory in which he criticizes the Utilitarian view of justice in favor of what he calls Teleological view of justice. Aristotle asks: if you have to distribute best flutes to who would you distribute? Generally, we come with responses like needy, interested or who can use best the flutes will deserve the best flutes. However, surprisingly, Aristotle denies these responses and says the best flutes have to be distributed to the best flute players who have the best excellence or virtues to practice the flute playing. Here, Aristotle offers his teleological explanation to this idea of justice that society is consists of different kinds of people having different virtues and people must be rewarded according to their best virtues by which they practice those virtues and merits at best to create a best society. Thus, justice in fair distribution has got a telos or purpose to give resources and opportunities to the people so that that can create a good society by practicing their virtues. Aristotelian view really throws light on a serious issue called reservation. A group of people who are suffering from the socio-economic equalities in education, job and business usually criticizes the other benefited groups and argue that there should not be inequalities be maintained by the governments through affirmative or positive actions. They further argue that due to reservation less qualified and skilled people get the powers and positions in governance and they produce poor quality results. But, this view is questionable on the grounds that the people of either group advantaged or

non-advantaged, meritorious or less qualified are doing the same thing corruption in the various systems without practicing their skills or virtues. As a result, the so called merit is not serving the purpose to create a better society and most of the people those who are getting opportunity and resources are utilizing and exploiting their powers and positions. Nevertheless, the reservation given in the name of cast and other arbitrary facts which are only by chance and not chosen by any individual is really poses a moral problem of forced discrimination. Therefore, the reservation policies must be in Rawlsian terms be based on qualified equality principle serving the least well-off to really improve the socio-economic condition of less advantaged people. The third prominent view offers a criticism of both the Rawlsian and Aristotelian views of justice. The third view advocates individual rights theory and of against active roles of government in fair distribution of resources. Immanuel Kant and Robert Nozick are the two main philosophers who actually defend the theory of rights and liberty of people. Immanuel Kant argues that individuals have rationality to decide their better off and thereby they must be given autonomy to choose their own ideas of good life. We should not lead by any purpose to create a better society; rather a better society will automatically come when the individuals will be given basic liberties and rights with a sense of autonomy. On the other hand, Robert Nozick rejects the active roles of governments and strong forms of affirmative action theories. He argues that government roles should be limited to the security and protection of people. There should not be any paternalistic legislation, moral legislation, and wealth re-distribution through taxation. It is in this way a government should not intervene in individuals' lives and should not limit their choices relating to education, job, business and life as a whole. This is what libertarians define as Minimal State Theory. The first two views and the third view in contrast get impression of a tussle between the Fit versus freedom. Rawls and Aristotle argue about the fitness of the principles and virtues, while Nozick and Kant defend the rights and freedom of individuals.

Only reservation is not the resolution and creamy layer must be excluded from the benefit of reservation. And, better executed reservation must not solving the purpose of cast-vote-politics, rather, it should purposefully lessen the socio-economic gaps amongst the people so that they can all compete in the race of development with ,if not an equal start then at least ,a fair start. Reservation in this way rationally be based on socio-economic real facts and conditions, and the socio-economic benefits should be distributed to those who really deserve without considering the arbitrary facts to which cast, class, race and gender they belong. Indeed we cannot really go in past to rectify what have done but taking meaningful lessons from the past it is better to give proper resources to all those who deserve

and want to compete and develop. It is good to create a better present than making it again an unjust past for some groups of people in coming future.

Questions:

9. What do you mean buy **the arbitrary facts**?
   a) They are about the cast, race and gender.
   b) The facts for which we cannot claim any moral credit.
   c) The random facts we do not choose.
   d) None of these.
10. Which of the following are the **right** points about the **Minimal State Theory?**
   a) Libertarians defend the theory.
   b) It objects moral legislation.
   c) It supports Individual freedom.
   d) It rejects paternalistic laws.
11. What is **moral desert**?
a) One should be rewarded for one's good actions and punished for the bad.
b) Justice should be based on one's talent & actual performance.
c) One should be respected on the grounds one actually deserves.
d) None of These.
12. Best Flutes must go to the best flute players. Which of the following statements are **false** according to this remark made by Aristotle?
a) Best flute players morally deserve best flutes.
b) Best flute players have interest in best flutes.
c) Best flute players have virtue to practice flutes at best.
d) Best flute players would use flutes at best.
13. Which of the following statements are **false** about the Rawls's thought experiment; veil of ignorance?
a) In veil of ignorance, individuals are psychologically fit and conscious about their self interests.
b) They are known to their arbitrary facts.
c) They know about their history.
d) It is their original position.

14. Which of the following statements are true about **Affirmative Action Theory?**

a) Some advantaged groups of people have to compensate for some least advantaged groups of people.

b) The different variety of groups who actually make up the whole society should be given proper and proportionate socio-political-economic representation.

c) A government should take positive actions to redress the wrongs of past.

d) None of These.

15. What are the basic points of **Teleological argument of justice**?

a) Society is consists of different kinds of people having different virtues.

b) People must be rewarded according to their best virtues.

c) A good society can be created by respecting the virtues.

d) Justice in fair distribution has got a telos or purpose to create a good society.

16. "Only reservation is not the solution". What does **author mean** to this statement?

a) Creamy layer must be excluded from the reservation.

b) Reservation must not a tool for cost-vote-politics.

c) Socio-economic benefits should not be distributed with the consideration of arbitrary facts.

d) Reservation should minimize the socio-economic gaps amongst the people.

17. What do you mean by the second principle of justice by John Rawls?

a) It is to manage the socio-economic differences for the least well-off.

b) It is the balance view of social justice.

c) It is the principle of difference (qualified equality).

d) None of these.

Section C

Passage: 3 (4X1=4)

What is the most demanding wish other than freedom? Indeed, it is respect for individuals' freedom. Theories of individual rights make the point clear that each individual has a separate life and different preferences. Individuals' preferences should not be sacrificed for either larger social purposes or the common good. We people not only desire but also value the freedom. By the very human nature we like freedom and do not want to live in the restrictions of laws and rules. Nevertheless, we desire those minimum laws and rules which actually protect

ourselves from others. Libertarians define freedom in such a way that freedom for individuals is all or nothing. In other words, they argue individuals must be given complete freedom. However, Libertarians differentiate between individual and collective (social) freedom. According to libertarians, there should not be any kind of limitations on individual freedom. However, there is a condition for collective freedom that one is free to the extent that one must not harm the other. Most interesting feature of the Libertarians' idea of freedom is that one has got complete freedom in a very precise sense that one can, if one wishes so, harm himself or herself. Libertarians advocate for the minimal state theory.

What this theory simply means is that state should not limit the liberty of individuals or a government should not regulate the individuals' choices and actions by propounding certain restrictions and laws. Individuals have got fundamental right to liberty. Government has got only role to play in individuals' lives, and that is to protect them from any kind of external threat like theft, fraud, and corruption. Libertarians object the excessive use of the powers of government in the way it intervene in individuals' lives and coerce them to follow certain laws. There are three such objections by Libertarians against the governments. First, government should not make paternalistic laws. Second, government should not make laws for the morals legislation of individuals' lives. Third, government should not redistribute wealth by putting taxes on individuals' salary. By paternalistic laws we mean such laws which are actually for the individuals' safety and protection like the helmet, seatbelt driving safety laws, and the pension laws. Libertarians argue that enforcing such laws on an individual a government undermines his or her risk taking ability. Libertarians believe that government should not propose a certain kind of values or principles for individuals' lives. An individual must be given full freedom to choose his or her own patterns and ways to live the life. However, Libertarians put one condition on this that one's moral choices must not affect the others. On the issue of taxation, Libertarians contend the government and say: by putting taxes on individuals' income government takes away the ownership of an individual. The amount we pay as taxes becomes slavery for an individual. What we earn through our labor talent or skill is our self expression. In this way, a self becomes slave and one's labor becomes forced labor. This is the reason governments should not redistribute the wealth by following the mechanism of taxation. However, governments can have a check on

individuals' property rights. A government must insure fair dealings amongst the individuals in terms of the initial holding of property, justice in acquisition and transfer of money, and other property rights. Libertarians by limiting the excessive role of government maintain: A government should respect an individual's personal life. The life of an individual must be respected as an end and never as means. We should respect the decisions of an individual and should not intervene in others decisions. Respect an individual's life projects independent of collective common goods. And, by respecting an individual a libertarian simply means: government should not interfere in individuals' personal life for accomplishing the purposes of society at large.

Questions:

18. Why do Libertarians object **Paternalistic laws**?
    a) Such laws undermine risk taking ability of an individual.
    b) Such laws restrict the individual freedom for protection.
    c) Such laws try to protect individuals and maintain that individuals should not misuse freedom and do not harm themselves.
    d) Such laws enhance individual freedom.
19. What is the **argument of Libertarians**?
    a) Individual freedom is all or nothing.
    b) Liberty is the fundamental right of an individual.
    c) Individual freedom is limited only in the sense that one is not free to harm other.
    d) An individual cannot harm oneself.
20. What does **respect mean** for a libertarian?
    a) An individual must be treated as an end.
    b) An individual must not be treated as a means to some other larger social purposes.
    c) Sometimes we can treat an individual as means.
    d) Individuals' decision making ability should be respected.
21. Why is **taxation wrong** according **to Robert Nozick**?
    a) One's profession becomes slavery.
    b) The amount we pay as tax is a kind of forced labor.
    c) It takes ownership of an individual.
    d) None of these.

Passage: 4 (1.5X10=15) (Attempt All Questions)

In the Kantian deontological system the centrality of the morality lies in the moral laws. Why it should not be the motive of engineering student to do engineering only for the job and job for the money? Following the same consequentiality logic why it should not be that students attend the lecture only for the attendance and not for the learning and the development of engineering guts and personality. It is not the purpose of engineering to acquire degree but is to create this world a better place to live rationally and freely.

The father of human rights explains the ideas of individual freedom and respect for human beings in so neat and scientific manner. Being contemporary to Sir Isaac Newton, Kant was very much influenced with Newtonian physics, and he was in full agreement with Sir Newton on the point that nature is governed by the deterministic physical laws. However, this determinism was a sort of challenge for Kant to define, that if we are governed by the deterministic laws then, in what sense we can be said to be free. Kant by respecting the Newtonian Physics initiates his argument in such a beautiful manner and says that there are two kinds of world: 1) Physical, and 2) Moral. Further, he argues that we people are governed by the physical laws in so far as we operate in nature (we operate under law of gravitation, in a certain atmosphere consisting of density, pressure and other physical conditions).

However, Kant argues that this is a limitation for the animal life because there exists the one and only physical world for them. But, for individual persons, there exists what Kant refers to as moral world. Moral world is such a world that we create for ourselves. And, we can create it by the virtue of human rationality. Moral world is a world of rationality in which the human life is governed by reasons.

Kant says it is only human beings who have got mind and its faculties of thinking and reasoning. By having recourse to these faculties of mind, an individual person can intend, resist and can do certain acts and can refrain herself/himself from doing something. Kant argues that a person is free, and free by reason. For example, let's compare two persons one is guided by reason and another who does not use rationality. You can

easily understand that the one who follows the reason can explain many events of life but the person who does not follow would be depending either on the other or would be always operating under certain fear or external command. This is the reason why Kant argues that it is rationality through which we get the freedom. Thus, Kantian idea of individual freedom, unlike the Libertarians, is grounded in human rationality. But Kant criticizes both the utilitarian and Libertarians. He argues that there is nothing wrong in attaining the pleasures but they should not be the sovereign masters to decide our CAD. On the other hand, he refuses the libertarians idea of respect and freedom that it is based on the fact that we own ourselves.

Human beings possess rationality and autonomy they must be respected because they have the capacity to legislate morality for themselves. They can create moral rules for their own lives. This is what Kant calls self-legislation. In other worlds, we can say one can realize the sense of freedom in creating rational rules for one's betterment of life. Kant further tries to explain that morality should be universal and uniform. It should not depend either on personal biasness or external pressure. Morality must be objectively rational in its character that we people can easily understand it without involving the personal reasons, and if we really want to include our personal reasons then we must not personalize the reasons but rationalize them to the extent that they become unbiased, universal and uniform.

Kant after explaining the ideas of rationality and freedom further moves to explain the idea of moral accountability and says that we people can be held responsible once we act freely and rationally. However, he makes a very interesting point that it is very much possible that a person does something 'X' and 'X' is right and s/he knows very well that 'X' is right but there is possibility that 'X' is not moral. Really interesting moral fact it is, Kant says that a person who does 'X' but not by his or her motive. The person may not have chosen that action or s/he was not free to do any other action. It means the person has performed the action either on the other's choice or under coercion, pressure or force. His own intention was not involved. The action may be right but in Kantian

definition it is not moral. So, you do something right but if not on your free intention then your action does not have any moral worth. It cannot be said morally a praiseworthy action.

Thus, we can praise or blame someone for his or her actions only in that someone has performed an action with freedom and reason. Kant says moral responsibility can be ascribed to someone who is both a free and rational creature. It is in this sense Kant emphasizes that morality should be free from religion. Kant says morality is an expression of moral law in oneself. Morality must freely evolve in oneself. Kantian description of morality is based on the deontology. Deontological morality is a framework in which moral worth of an action, duty, rule, decision or person lies in the very nature of the action, rule, duty, decision or person.

Unlike Utilitarianism, Kantianism focuses on the motive, sense of duty and nature of an action. We should be careful to the means that we choose to fulfill an end and there is no need to look at results. Kant says that even if your right means do not produce the right end do not mind. This view holds we should not get depressed as your action carry a moral worth within itself. For example, suppose if someone being student honestly study hard by practicing his duty somehow does not get good marks, then this student, according to Kant, need not be worried and disappointed in so far as his action is morally correct and good. Kant put it in this way "**Duty for the sake of duty.**"

Kant maintains universality in his morality by explaining the idea of categorical imperative. He argues in all moral rules there is an imperative but it can be of two kinds: 1) Hypothetical, and 2) Categorical. Hypothetical imperative means the imperative or moral command of an action depends on external conditions. For example, if you want to get good marks then study hard. In this example for a student studding hard is a morally correct thing but it has to qualify a condition, and it is to score the good marks. Categorical imperative is supreme principle of morality. However, categorical imperative means the imperative or moral command of an action is unconditional. For example, do your duty. Kant argues the moral worth of a moral action

does not depend on the consequences it produces but depends on our intentions behind our actions. For example, a murderer and a doctor both uses a knife but with different intentions one uses to kill a life and other uses to save a life. Here Kant introduces his beautiful idea of good- will and says that no action can be said to have moral worth unless performed by a sense of good will. What is good-will? Kant replies this question so neatly and says that a will is different from desire. Desires are psychological and represent our emotions and passions. And, they lack determination that is why our desires get changing all the time. However, will is not purely psychological, it is based on reasons and have got determination as well. Kant says, although will is rational and has got strong determination but without being good a normal will may carry wrong intentions. For example; a strong willed person may use his determination in killing someone. This is the reason why Kant argues that our will must follow good intentions that's what he calls good-will. It is a form of categorical imperative. He says without good-will morality cannot be performed and all values say knowledge, power and beauty can be misused. For example, a knowledgeable person can misuse the knowledge for the wrong purposes making cheat and frauds, powerful person can misuse the power by beating a week. Without having good-will we cannot be moral in a universal, uniform manner and cannot perform action in the sense of moral necessity. Kant writes: Two things fill the mind with ever new and increasing admiration and awe, the more often and steadily reflection is occupied with them: *the starry heaven above me and the moral law within me.* One is the expression of natural universality, uniformity and necessity. On the other, it is the expression of individual freedom, and universal uniform and morally necessity within oneself.

Questions:

22. Why does Kant criticize the utilitarian and libertarian grounds for CAD?
    a) CAD cannot be based on pleasure and pain.
    b) CAD should not be based on what we desire.
    c) Our freedom is not based on our ownership.
    d) Morality is not grounded in our pleasurable experiences

23. Which of the following statements are **correct** about the moral world?
   a) It is physical world.
   b) It is the world where universality, freedom and moral necessity are possible altogether.
   c) It is the world where determined laws work.
   d) It is an intelligible world where rationality works.
24. "Humans are rational & autonomous beings". What does **Kant mean** by the statement?
   a) They have capacities to decide on CAD by reasons.
   b) They can create moral laws for themselves.
   c) They have rationality to exercise their freedom.
   d) They are free from laws of nature.

25. "What is **humanity principal** according to Kant?
   a) All human beings have autonomous life.
   b) They can rationally practice morality.
   c) They are rational beings and end in themselves.
   d) They are means like objects and other thing.
26. We **cannot** hold someone morally responsible only when:
   a) S/he is not free to perform an action.
   b) S/he has chosen the action by his or her own will.
   c) S/he did a right thing on other's motive.
   d) S/he is irrational being.
27. **"The starry heaven above me and moral law within me".** What Kant means to this statement?
   a) An expression of moral law in oneself.
   b) Moral law within me is governed by deterministic laws of nature.
   c) It is the Universal morality within me by moral laws as present in the universe by laws of nature.
   d) It is universal uniform and moral certainty in one's moral behavior.
28. What is **right** about the Good-will?
   a) Good-will carries good desires.
   b) Good-will is the expression of categorical imperative.
   c) Good-will carries good intentions.
   d) Good-will is governed by physical satisfaction.

29. What is **supreme principle of morality** and why?
   a) Categorical imperative is a supreme principle of morality.
   b) Because moral command in categorical imperative will bring good-will and universality.
   c) Because the moral command is chosen by God.
   d) Because moral command in categorical imperative has nothing to do with the consequence, result or end of an action.

30. Which of the following statement is/ are **correct** for deontological morality framework?
   a) Moral worth of an action lies in the result of an action.
   b) Moral worth of an action lies in the good-will behind the actions.
   c) Moral worth of an action lies in the will to achieve self interests.
   d) Moral worth of an action lies in motive of an actor.

31. A person 'P' performs an action 'X'. And, 'X' is right. And, P knows that 'X' is right. But, his action 'X' can be said to be **moral only if**:
   a) P was not free to perform 'X'.
   b) 'X' is his/her free choice.
   c) 'X' was a rational choice for P.
   d) P performed 'X' on his own intention.